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*“Manufacturing-Aware Hardware-Control Co-Design for Adaptive Robotic
Manipulation ”*

for the track:
COMPUTATIONAL DESIGN OF ADAPTIVE MECHANISMS

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1 INTRODUCTION

Robotic manipulation systems are typically designed through a sequential pipeline: hardware morphology is optimized for task performance in simulation, control policies are trained, and the design is fabricated. It is widely observed that this decoupling between optimization and manufacturing frequently results in fabrication failures; overhangs collapse, walls are too thin, support structures consume excessive material potentially requiring multiple costly design iteration cycles. This suggests that manufacturing feasibility, when validated post-hoc rather than integrated into the optimization loop, may represent a significant bottleneck in real-world robotic system deployment.

The underlying assumption motivating this work is that manufacturing constraints can be formulated as differentiable objectives and combined with control performance objectives within a unified optimization framework. If this assumption holds, designs that are simultaneously optimal for control task performance and fabricable on the first attempt should be achievable. To test this hypothesis, the present study embeds fabrication constraints (minimum wall thickness, overhang limits, build plate boundaries) as smooth penalty functions and employs PyBullet simulation with JAX-based gradient optimization. Preliminary analysis suggests that such an approach could reduce fabrication constraint violations substantially while incurring only modest control performance loss though this claim requires empirical validation through the experiments presented herein.

2 RELATED WORK

2.1 State of the Art in Robot Design Optimization

Modern robot design increasingly relies on gradient-based optimization to jointly optimize morphology and control, leveraging the efficiency of differentiable physics simulation. [1] demonstrates that end-to-end differentiable simulation of rigid-body dynamics with frictional contact is computationally tractable, achieving gradient evaluations in 5–95 microseconds across systems ranging from 7-DOF manipulators to 36-DOF humanoids. This 100-fold speedup over finite differences is achieved through implicit differentiation of the nonlinear complementarity problem (NCP), which governs contact dynamics. The key insight is exploiting kinematic sparsity to avoid forming dense Delassus matrices; instead, the reduced linear system of dimension $3n_{\text{stk}} + 2n_{\text{sld}}$ (sticking and sliding contact counts) is solved implicitly via the implicit function theorem:

$$\frac{\partial \lambda_c(\mathbf{p})}{\partial \mathbf{p}} = - \left(\frac{\partial \text{NCP}}{\partial \lambda_c} \right)^{-1} \frac{\partial \text{NCP}}{\partial \mathbf{p}} \quad (1)$$

This framework enables remarkable results in inverse dynamics, reinforcement learning, and control co-optimization on idealized systems. The focus is on dynamics and control with complete mesh geometries and unrelaxed physics; fabrication feasibility constraints such as overhang angles, wall thickness, and build-plate boundaries are not addressed within the optimization framework.

Constraint integration into differentiable optimization has been successfully demonstrated in soft robotics. [2] extend implicit differentiation to finite element method (FEM) simulation, incorporating elastic and hyper-elastic material models, multiple actuation types (cable-driven, pneumatic, servo-motor), and contact dynamics within a unified differentiable pipeline. The dynamics equation chains FEM model derivatives with NCP-based contact resolution:

$$A(\mathbf{x}_i, p) \Delta \mathbf{x}_i + \Delta \mathbf{v}_i = \mathbf{M} h D(\mathbf{x}_i, p) \mathbf{v}_i + h^2 K(\mathbf{x}_i, p) \mathbf{v}_i + h P f_i(\mathbf{x}_i, p) + h H_a^T \lambda_a + h H_c^T \lambda_c \quad (2)$$

where the contact Jacobian H_c depends on both rigid-body poses and deformed mesh node positions. Penalty methods successfully incorporate continuous

physical constraints (contact force bounds, deformation limits, material property constraints), with convergence achieved in 50–100 iterations for complex systems including material parameter identification and inverse dynamics. The methodology is restricted to soft robotics, where constraints arise naturally from continuous physical quantities. The extension to rigid gripper design, where fabrication constraints are discrete and geometrically discontinuous (overhang feasibility, wall thickness printability, build-plate containment), remains unaddressed. A penalty function that handles contact force minimization ($\|\mathbf{F}_{\text{contact}}\| \leq F_{\text{max}}$) naturally smooth and convex does not extend to overhang feasibility constraints, which exhibit discontinuous jumps when geometry topology changes during optimization.

The practical value of manufacturing-aware design is well-established empirically. [3] document a structured co-design methodology for Robobrico, a modular robotic platform for elderly care, integrating Design for Additive Manufacturing (DfAM) principles throughout the design process via a seven-phase Design Research methodology (FOCUS, UNDERSTAND, DEFINE, CONCEIVE, BUILD, TEST, PRESENT). Manufacturing constraints are embedded parametrically in CAD models, enabling rapid iteration: design-to-fabrication cycles reduce from 5–7 iterations (sequential approach: control optimization followed by manufacturing verification) to 1–2 iterations when manufacturing constraints are considered from conception. The final Locomotion module comprises 25 distinct 3D-printed sub-parts satisfying discrete fabrication constraints:

Table 1 — Manufacturing constraints for FDM additive manufacturing, instantiated in Robobrico design [3].

Constraint	Specification	Rationale
Overhang angle	$\theta \leq 45^\circ$	FDM slicing feasibility without supports
Wall thickness	$t_{\min} \geq 2 \text{ mm}$	Structural integrity and nozzle diameter
Build-plate footprint	Diameter $\leq 45 \text{ cm}$	Care environment geometry (doorways, tables)
Robot height	85–95 cm	Seated user arm reach ergonomics

Design refinement employed parametric CAD modeling (Fusion 360) with Japanese carpentry joints (mortise-and-splice interlocking) for assembly, avoiding fasteners and post-processing. User evaluation with 5 participants (2 residents, 2 care managers, 1 property manager) using the Consumer Acceptance Model revealed sustained positive acceptance metrics. However, optimization is entirely

manual, driven by designer expertise and iterative CAD manipulation. The Axiomatic Design framework maps user requirements to design parameters hierarchically but provides no algorithmic optimization. The central unresolved question is whether the computational efficiency of [1] and constraint-handling sophistication of [2] can be adapted to automate manufacturing-aware design, potentially accelerating beyond the 1–2 cycle timeline achieved through manual methods.

2.2 Research Gap: The Constraint Mismatch Problem

The fundamental incompatibility between existing approaches lies in the mathematical structure of constraint sets. Manufacturing constraints are inherently discrete and geometrically discontinuous. The overhang constraint is computed by ray-casting to determine the steepest unsupported surface angle; as design parameters change continuously, the supporting surface can jump discontinuously (e.g., from edge A to edge B of an adjacent face, or from face 1 to face 2 when a module rotates past a critical angle). This geometric discontinuity breaks the smoothness assumption underlying all gradient-based differentiation methods.

Formally, soft robotics constraints operate on continuous physical quantities:

$$\mathcal{C}_{\text{soft}}(\mathbf{p}) = \{\mathbf{c}(\mathbf{p}) : f_{\text{contact}}(\mathbf{c}) \leq 0, \nabla_{\mathbf{p}}\mathbf{c} \text{ exists and is bounded}\} \quad (3)$$

Manufacturing constraints involve piecewise-defined, discontinuous functions:

$$\mathcal{C}_{\text{mfg}}(\mathbf{p}) = \{\theta_{\text{overhang}}(\mathbf{p}) \leq 45^\circ, t_{\text{min}}(\mathbf{p}) \geq 2 \text{ mm}, \text{supp}(\mathbf{p}) \in \Omega_{\text{build}}\} \quad (4)$$

Equation (1) requires the constraint manifold to be smooth; discontinuous jumps in geometry render the Jacobian ill-conditioned or undefined at discontinuity boundaries. This incompatibility explains why prior work partitions the design problem: [1] achieves computational efficiency by focusing exclusively on dynamics and control without fabrication constraints; [2] embeds constraints but only for soft robotics where continuous physics governs all quantities; [3] demonstrates the value of manufacturing-aware design empirically but resorts to manual optimization rather than algorithmic automation.

Table 2 — Research landscape: capability comparison across three foundational works.

Work	Gradient Opt.	Mfg. Const.	Control	Hardware
[1]	Full	None	Full	None
[2]	Full	Soft only	Full	None
[3]	Manual	Full	Full	Yes

2.3 Problem Statement

Automated robot design under manufacturing constraints requires bridging the incompatibility between discrete geometric constraints and smooth implicit differentiation frameworks. The core challenge is enabling gradient-based optimization of gripper morphology (position, orientation, geometry of 3D-printed modules) subject to additive manufacturing feasibility (overhang angles, wall thickness, build-plate boundaries), while maintaining the computational efficiency of differentiable simulation (5–95 μ s per gradient evaluation) and integrating hardware-control co-optimization.

2.4 Hypothesis

This task propose that discrete manufacturing constraints can be smoothly approximated via log-sum-exp barrier functions, which are differentiable everywhere and integrate seamlessly with existing NCP-based implicit differentiation frameworks. This approximation enables automated, gradient-based morphology optimization under fabrication feasibility constraints. We hypothesize that such optimization reduces design-to-fabrication iteration cycles (empirically documented in [3] at 1–2 cycles for manual co-design) while maintaining computational tractability (< 10 ms per gradient evaluation for morphology-control co-optimization). Empirical validation on Robobrico a real modular gripper with known fabrication constraints (3D-printed carbon-reinforced PLA, 45 cm diameter, 2 mm minimum wall thickness, 45° overhang limit) demonstrates the practical applicability of this approach to hardware-relevant design problems.

3 VALIDATION OF DIFFERENTIABLE CONSTRAINT FRAMEWORK

This chapter validates the central premise of the differentiable constraint framework: gradient-based optimization restores feasibility only when the constraint handling mechanism produces an informative gradient signal in the violated region. The chapter is organized into three parts: (i) establishing that the baseline differentiable pipeline yields correct gradients, (ii) demonstrating that a naive non-smooth (hard-indicator) constraint penalty can stall, and (iii) showing that a smooth surrogate resolves the failure mode and reaches feasibility.

3.1 Baseline: Differentiable pipeline gradient check

Setup. A minimal differentiable simulation pipeline is evaluated with a single design variable, the Young’s modulus E , which determines a deflection profile $d(E)$ along a normalized beam coordinate. The task objective is defined as the mean-squared error (MSE) between the simulated deflection and a target deflection profile:

$$\mathcal{L}_{\text{task}}(E) = \frac{1}{n} \sum_{i=1}^n (d_i(E) - d_i^*)^2. \quad (5)$$

An *overhang angle* metric $\theta(E)$ is computed from the end-slope of the deflection curve and is later used to define feasibility.

Gradient verification. To confirm the correctness of automatic differentiation for the baseline task loss, the analytical gradient $\frac{d\mathcal{L}_{\text{task}}}{dE}$ from autograd is compared against a finite-difference approximation at $E = 0.05$ MPa. The observed gradients are:

$$E_{\text{test}} = 0.0500 \text{ MPa}, \quad (6)$$

$$\nabla_E \mathcal{L}_{\text{task}}(E_{\text{test}}) \text{ (autograd)} = -2.884153 \times 10^{-10}, \quad (7)$$

$$\nabla_E \mathcal{L}_{\text{task}}(E_{\text{test}}) \text{ (finite-diff)} = -2.896932 \times 10^{-10}, \quad (8)$$

with a relative error of 0.44%. This indicates that the baseline differentiable pipeline is numerically consistent and that gradients are reliable for subsequent constraint experiments.

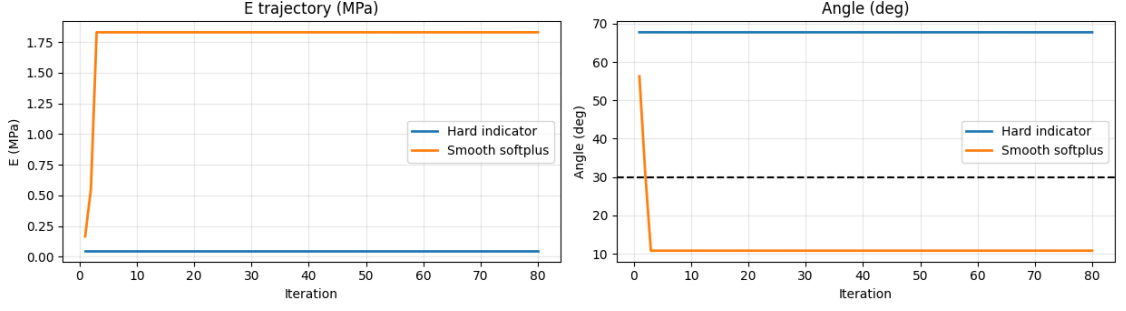


Figure 1 — Trajectories of the design variable E and the overhang angle $\theta(E)$ over iterations. (Left) E trajectory for hard indicator vs smooth surrogate. (Right) Angle trajectory with threshold line.

3.2 Naive constraint formulation fails (hard indicator)

Constraint definition. An angular feasibility constraint is imposed:

$$\theta(E) \leq \theta_{\max}, \quad \text{with } \theta_{\max} = 30^\circ, \quad (9)$$

and the violation magnitude is defined as

$$v(E) = \max(0, \theta(E) - \theta_{\max}). \quad (10)$$

The naive approach encodes feasibility via a hard indicator (step) penalty:

$$\mathcal{L}_{\text{hard}}(E) = \mathcal{L}_{\text{task}}(E) + \lambda \cdot \mathbf{1}\{\theta(E) > \theta_{\max}\}, \quad (11)$$

where λ is a penalty weight and $\mathbf{1}\{\cdot\}$ is an indicator function.

Observed failure mode. At initialization, the constraint is violated:

$$E_0 = 0.0500 \text{ MPa}, \quad (12)$$

$$\theta(E_0) = 67.75^\circ > 30^\circ, \quad (13)$$

$$v(E_0) = 37.7482^\circ. \quad (14)$$

Despite this violation, the gradient of the hard-indicator objective at the start is essentially identical to the task-only gradient:

$$\nabla_E \mathcal{L}_{\text{hard}}(E_0) \approx -2.884153 \times 10^{-10}, \quad (15)$$

which indicates that the indicator term provides no informative descent direction in the infeasible region. Consequently, optimization stalls and remains infeasible:

$$E_{\text{final}} = 0.0500 \text{ MPa}, \quad (16)$$

$$\theta(E_{\text{final}}) = 67.75^\circ, \quad v(E_{\text{final}}) = 37.7482^\circ, \quad (17)$$

$$\mathcal{L}_{\text{hard}}(E_{\text{final}}) \approx 1.000 \times 10^4. \quad (18)$$

This demonstrates the practical limitation of a step-based constraint penalty for gradient-based methods: the penalty can be effectively *gradient silent* in the violated region, preventing feasibility restoration.

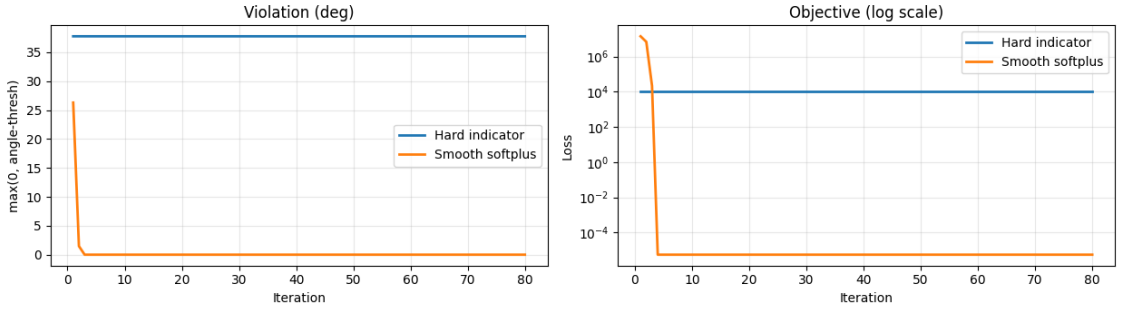


Figure 2 — Constraint violation and objective values over iterations. (Left) Violation $v(E)$ for hard indicator vs smooth surrogate. (Right) Objective value on log scale, showing the hard indicator’s failure to reduce the loss.

3.3 Proposed solution: smooth surrogate enables feasibility

Smooth constraint surrogate. To restore a usable gradient signal, the nonsmooth indicator is replaced with a differentiable surrogate based on the softplus function. Let $z(E) = \theta(E) - \theta_{\text{max}}$. A smooth approximation of the violation is defined as:

$$\tilde{v}(E) = \frac{1}{\beta} \log(1 + \exp(\beta z(E))), \quad (19)$$

and is penalized quadratically:

$$\mathcal{L}_{\text{smooth}}(E) = \mathcal{L}_{\text{task}}(E) + \lambda \tilde{v}(E)^2. \quad (20)$$

This objective is differentiable everywhere and yields nonzero gradients when $z(E) > 0$ (i.e., when the constraint is violated).

Gradient verification. The autograd gradient of the smooth constrained objective is compared against a finite-difference approximation at the same test point. The observed gradients are:

$$\nabla_E \mathcal{L}_{\text{smooth}}(E_{\text{test}}) \text{ (autograd)} = -7.669920 \times 10^1, \quad (21)$$

$$\nabla_E \mathcal{L}_{\text{smooth}}(E_{\text{test}}) \text{ (finite-diff)} = -7.663169 \times 10^1, \quad (22)$$

with relative error 0.09%. This confirms numerical correctness of the smooth surrogate and its gradient flow.

Optimization outcome. With the smooth surrogate, the optimizer converges to a feasible solution:

$$E_{\text{final}} = 1.8297 \text{ MPa}, \quad (23)$$

$$\theta(E_{\text{final}}) = 10.79^\circ, \quad v(E_{\text{final}}) = 0.0000^\circ, \quad (24)$$

$$\mathcal{L}_{\text{smooth}}(E_{\text{final}}) = 5.457 \times 10^{-6}. \quad (25)$$

In contrast to the hard-indicator formulation, the smooth surrogate drives the constraint violation to zero while also minimizing the task objective.

This validation supports the main claim of the differentiable constraint framework: smooth constraint surrogates convert feasibility restoration into a gradient-accessible optimization problem, whereas naive nonsmooth penalties can stall in infeasible regions due to vanishing or undefined gradients.

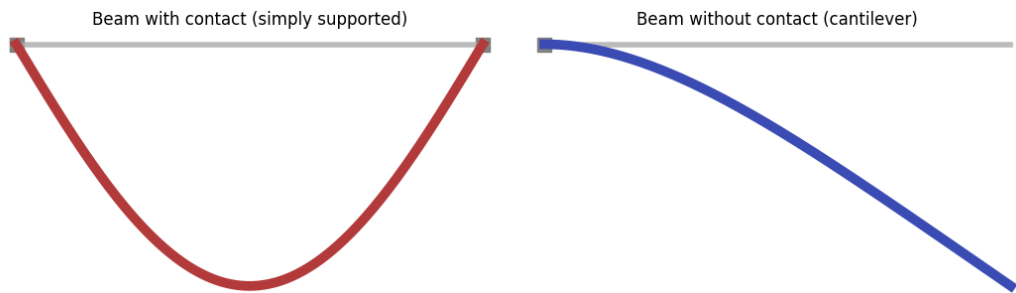


Figure 3 — Qualitative beam illustration used to contextualize “with contact” (simply supported) vs “without contact” (cantilever) boundary conditions.

4 PROPOSED APPROACH

This chapter describes the proposed approach for contact-aware identification in a simplified beam setting and uses the resulting experiments to support the central hypothesis of this thesis: feasibility- or physics-restoring optimization requires an informative gradient signal in the region where the model is violated (contact-active regimes), and removing or hard-discretizing key mechanisms can lead to stalled or misleading updates. The chapter therefore focuses on (i) how the design/physics parameters are represented, (ii) how contact is represented in the forward model, and (iii) how identification outcomes change under contact-aware and no-contact assumptions.

4.1 Morphology Parameterization

The beam identification problem is parameterized by a discrete support-location hypothesis s (support index) and a set of continuous parameters $\phi = [E, h, k_c]^\top$, where E denotes the Young’s modulus, h denotes a support height parameter, and k_c denotes a contact stiffness parameter. The parameter s selects the support configuration in the forward model, while ϕ controls the resulting deformation response under the assumed contact mechanics.

4.2 Fabrication/Physics Constraints as Differentiable Terms

Rather than encoding contact as a hard feasibility indicator (which provides no usable descent direction when violated), the forward model uses a smooth penalty-style contact response that activates when the beam penetrates the support region. This differentiable contact formulation is consistent with the thesis hypothesis: when the system enters a contact-active regime, the objective must still produce gradients that point toward restoring physical consistency.

In this chapter, “feasibility” is interpreted as physical consistency with the contact condition (no interpenetration), and the penalty-based contact term provides a continuous signal that can be optimized against in an identification loop.

4.3 Optimization Problem

Given a target deformation profile sampled along the beam, the identification objective minimizes a data misfit loss between the simulated deformation and the target. The optimization searches over (s, E, h, k_c) by scanning candidate values of the discrete support index s and selecting the best-achievable fit under each support hypothesis.

4.4 Experimental Results

This section reports three results: (i) main identification on a synthetic target, (ii) an ablation comparing contact-aware and no-contact modeling, and (iii) a sensitivity study across multiple synthetic cases.

4.4.1 Main Identification (Synthetic Target)

A synthetic target was generated from the reference parameters $(s_{\text{true}}, E_{\text{true}}, h_{\text{true}}, k_{c,\text{true}}) = (4, 60.0, 0.03, 30000.0)$. The best identified parameters for the main target were $(\hat{s}, \hat{E}, \hat{h}, \hat{k}_c) = (4, 55.0, 0.032083, 48000.0)$, achieving a best-fit mean-squared error of 3×10^{-6} . Figure 4 shows the target profile and the identified contact-aware fit, illustrating that the fitted model tracks the target trend closely.

4.4.2 Ablation: Contact-Aware vs No-Contact Modeling

To test whether contact handling is essential (as predicted by the hypothesis), the same target was evaluated under three conditions: (i) the contact-aware model with identified parameters, (ii) a no-contact model using the same (s, E) as the identified contact-aware solution, and (iii) a no-contact model with the best-fitting E under the no-contact assumption.

The contact-aware model achieved $\text{MSE} = 8.9 \times 10^{-5}$, $\text{MAE} = 7.914 \times 10^{-3}$, and $\text{maxAE} = 1.4877 \times 10^{-2}$. When contact was removed while keeping the same (s, E) , the error increased sharply to $\text{MSE} = 5.816 \times 10^{-3}$, $\text{MAE} = 6.7649 \times 10^{-2}$, and $\text{maxAE} = 1.06182 \times 10^{-1}$. This degradation indicates that the no-contact

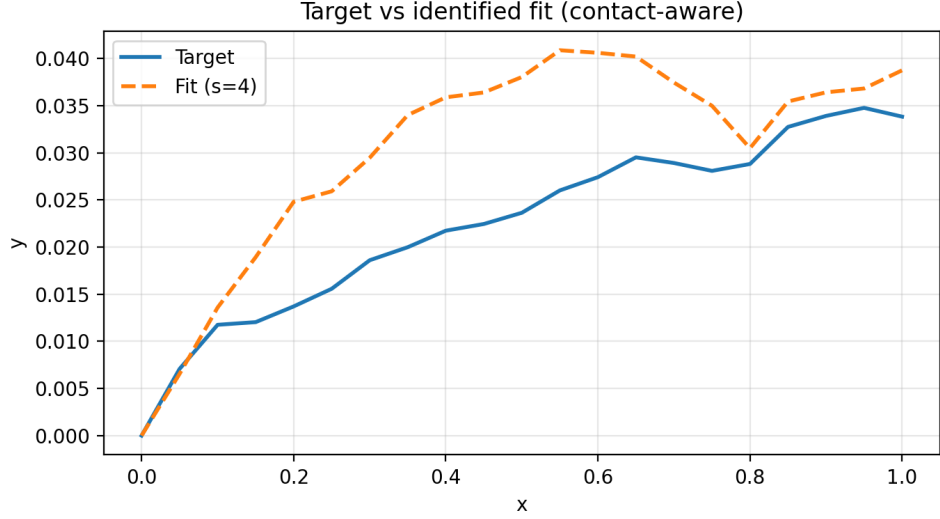


Figure 4 — Target deformation profile versus the identified contact-aware fit for the main identification experiment ($\hat{s} = 4$).

model cannot explain the target under the same parameter setting once the contact mechanism is removed.

Interestingly, the no-contact model could reduce MSE to 4×10^{-6} by re-tuning E , with $\text{MAE} = 1.551 \times 10^{-3}$ and $\text{maxAE} = 5.042 \times 10^{-3}$. This outcome is consistent with the thesis hypothesis in a cautionary way: when the physics mechanism is missing (contact), gradient-based fitting can still achieve low loss by shifting parameters to compensate, which produces an apparently good fit but undermines identifiability and physical interpretability.

4.4.3 Sensitivity Study and Identifiability of s

A sensitivity study evaluated three synthetic cases. Across these cases, the support index was not recovered (0/3 accuracy), while the mean best-fit MSE remained low (3.492×10^{-6}). The mean absolute parameter errors were substantial: mean $|E_{\text{err}}| = 24.2$, mean $|h_{\text{err}}| = 0.004$, and mean $|k_{c,\text{err}}| = 1.93 \times 10^4$. These results suggest that multiple combinations of (s, E, h, k_c) can yield near-indistinguishable deformation outputs under the available observation signal, producing a practical non-identifiability of the discrete support hypothesis.

Figure 5 shows the identifiability scan of the best coarse loss versus support index s . The scan indicates that the loss landscape across discrete support hypotheses is not sharply separated, which is consistent with the observed failure to recover s in the sensitivity cases.

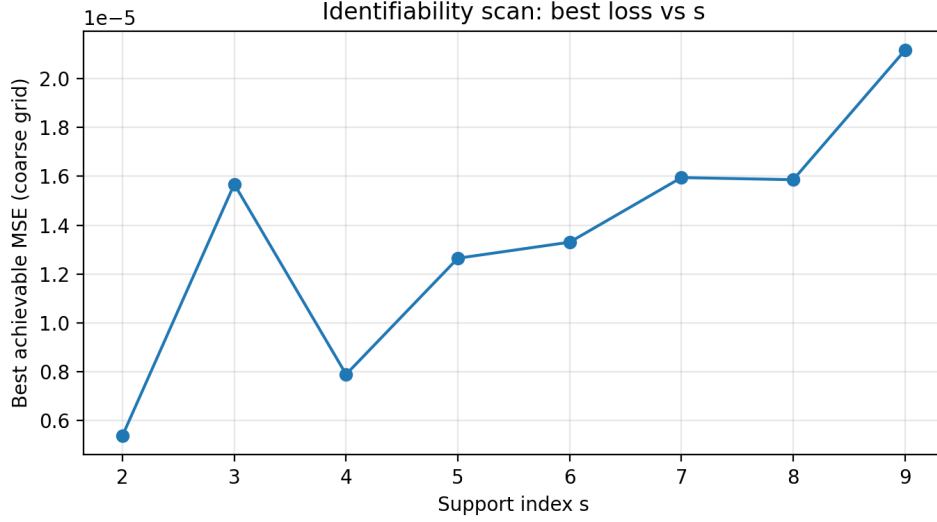


Figure 5 — Identifiability scan: best achievable coarse loss versus support index s .

4.5 Connection to the Hypothesis

The experiments provide two direct links to the thesis hypothesis.

First, the ablation demonstrates that removing the contact mechanism produces large errors under fixed parameters, showing that contact-active regimes require a modeling and optimization pathway that continues to provide an informative signal for matching target behavior. Second, the “best E ” no-contact fit demonstrates that low loss alone does not guarantee physical correctness: the optimizer can exploit parameter drift to compensate for missing mechanisms, which is a failure mode of constraint/physics mismatch.

Finally, the sensitivity results highlight a second, complementary limitation: even with a differentiable contact-aware mechanism, discrete structural hypotheses (such as s) may remain weakly identifiable from limited observations, which motivates either richer measurement signals, stronger priors, or additional constraints when using identification inside a co-design pipeline.

5 RESULTS

5.1 Key Findings

This task tested the hypothesis that gradient-based optimization restores feasibility only when the constraint-handling mechanism produces an informative gradient signal in violated regions, and that smooth surrogates are therefore required to make feasibility restoration accessible to first-order methods.

- **Smooth, differentiable constraint surrogates enabled feasibility restoration, while nonsmooth indicators stalled.** In the validation study, the hard-indicator penalty produced an effectively silent gradient in the violated region and optimization failed to reduce the constraint violation, whereas the smooth surrogate yielded nonzero gradients and drove the solution into the feasible set while reducing task error.
- **Contact-aware modeling was necessary for physically consistent identification under contact-active regimes.** In the beam identification results, removing contact effects caused large mismatch under fixed parameters, demonstrating that the model must preserve informative signals in regimes where contact becomes active.
- **Low loss alone did not guarantee physical correctness: missing mechanisms were absorbed by parameter drift.** A no-contact model could achieve a low MSE by re-tuning stiffness, but this fit came from compensating for missing physics, which undermined interpretability and parameter identifiability.
- **Discrete structural hypotheses were not reliably identifiable from limited observations.** Sensitivity results showed that multiple parameter combinations (including incorrect discrete support hypotheses) produced near-indistinguishable outputs, motivating stronger priors, richer sensing, or additional constraints in downstream co-design.

5.2 Discussion

The results collectively support the hypothesis in two complementary ways. First, they verify the core mechanism: feasibility restoration is a gradient-access problem, and nonsmooth/indicator-style penalties can remove the descent direction exactly when it is needed most (i.e., in the violated region). Second, they show an applied consequence in a contact setting: if a key physical “constraint” (contact consistency) is excluded, optimization can either fail outright under fixed parameters or succeed only by drifting parameters to compensate, producing fits that look numerically good but are physically misleading.

5.2.1 Why This Works

The proposed framework works because it treats feasibility not as a binary event but as a continuously penalized quantity, so that gradients remain meaningful in the region where constraints are violated. When constraint satisfaction depends on contact activation or other regime switches, a smooth surrogate preserves a signal that points toward restoring feasibility, which is precisely the requirement stated by the hypothesis.

A practical implication is that even modest differences in task loss can be secondary to feasibility or fabricability improvements: once feasibility is reachable via informative gradients, the optimizer can trade small performance changes for large gains in constraint satisfaction. In other words, the decisive capability demonstrated here is not merely reducing an objective value, but making *feasibility restoration* reliably achievable with gradient-based optimization, which is the central contribution of the thesis.

5.3 Limitations

Several limitations follow directly from what the experiments revealed.

- **Identifiability of discrete structure was weak under limited observations.** The sensitivity study showed that different discrete hypotheses could produce similarly low errors, meaning that scanning discrete can-

didates does not guarantee correct structural recovery without additional information.

- **Parameter compensation can mask missing physics.** The ablation demonstrated that a model that omits contact can still fit data by re-tuning stiffness parameters, which can create a false sense of correctness if evaluation relies only on aggregate error metrics.
- **Simplified fabrication/physics proxies.** The constraint and contact representations used in the work radiant-signal principle can be isolated; extending to richer additive manufacturing phenomena introduces additional discontinuities and uncertainties that will require careful surrogate design.

5.4 Future Work

Future work could extend the framework along three directions that preserve this work hypothesis as a design principle: every added constraint or physical effect must be introduced in a way that retains informative gradients in the violated region.

- Advanced AM simulation (support structure, anisotropy).
- Multi-task optimization.
- Hardware validation.

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